

HKCEE 1980

Mathematics II

80
1. $2ab - a^2 - b^2 =$

- A. $(a - b)^2$
 B. $(-a - b)^2$
 C. $(-a + b)^2$
 D. $-(a + b)^2$
 E. $-(a - b)^2$

80
2. $125^a \cdot 5^b =$

- A. 625^{a+b}
 B. 625^{ab}
 C. 125^{a+3b}
 D. 5^{a+3b}
 E. 5^{3a+b}

80
3. If $4p = 9q$, then $\frac{4p^2}{9q^2} =$

- A. 1
 B. $\frac{4}{9}$
 C. $\frac{9}{4}$
 D. $(\frac{4}{9})^2$
 E. $(\frac{9}{4})^2$

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4. If $n = 10^a$, then $\log_{10} n =$

- A. 10^a
 B. 10^n
 C. n^a
 D. a^n
 E. a

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5. $\frac{x^{-2} - y^{-2}}{x^{-1} - y^{-1}} =$

- A. $x^{-1} + y^{-1}$
 B. $x^{-1} - y^{-1}$
 C. $x^{-3} - y^{-3}$
 D. $\frac{1}{x - y}$
 E. $\frac{1}{x + y}$

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6. If $\frac{1}{x} = a + b$ and $\frac{1}{y} = a - b$,
then $x + y =$

- A. $\frac{2}{a}$
 B. $\frac{a^2 - b^2}{a}$
 C. $-\frac{a^2 - b^2}{a}$
 D. $\frac{2a}{a^2 - b^2}$
 E. $\frac{-2a}{a^2 - b^2}$

80
7. If $x = \frac{y + (n-1)z}{n+1}$, then $n =$

- A. $\frac{x - y + z}{z}$
 B. $\frac{x + y - z}{z}$
 C. $\frac{y - x - z}{x + z}$
 D. $\frac{y - x - z}{x - z}$
 E. $\frac{y + x - z}{x - z}$

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8. $\frac{5^{n+2} - 35(5^{n-1})}{18(5^{n+1})} =$

- A. $\frac{1}{18}$
- B. $\frac{1}{15}$
- C. $\frac{1}{5}$
- D. 5
- E. 5^n

80 Solve the inequality

9. $(4x + 3)(x - 4) > 0$

- A. $x > 4$
- B. $4 > x > -\frac{3}{4}$
- C. $-\frac{3}{4} > x$
- D. $-\frac{3}{4} > x$ or $x > 4$
- E. $x > -\frac{3}{4}$

80 When the hour hand has turned through an angle x° , what is the angle through which the minute hand has turned?

- A. $6x^\circ$
- B. $12x^\circ$
- C. $60x^\circ$
- D. $360x^\circ$
- E. $3\ 600x^\circ$

80 The first term of an arithmetic progression is 6 and its tenth term is three times its second term. The common difference is

- A. 18
- B. 4
- C. 3
- D. 2
- E. 1

80 A man sold a car for \$35 000 at a loss of 30% on the cost price. What would have been the loss or gain percent if he had sold it for \$50 500?

- A. A gain of 10%
- B. A gain of 1%
- C. No gain nor loss
- D. A loss of 10%
- E. A loss of 1%

80 If the length of a rectangle is increased by 10% and the width decreased by 10%, which of the following is true?

- A. Its area remains the same
- B. Its area is decreased by 1%
- C. Its area is increased by 1%
- D. Its area is decreased by 10%
- E. Its area is increased by 10%

80 The length of a side of a rhombus is 10 cm. If its shorter diagonal is of length 12 cm, what is the area of the rhombus in cm^2 ?

- A. 60
- B. 96
- C. 100
- D. 120
- E. 192

80 If the bearing of B from A is $\text{S}30^\circ\text{W}$, then the bearing of A from B is

- A. $\text{N}30^\circ\text{E}$
- B. $\text{N}60^\circ\text{W}$
- C. $\text{N}60^\circ\text{E}$
- D. $\text{S}30^\circ\text{W}$
- E. $\text{S}30^\circ\text{E}$

80 16. $\frac{1}{\frac{1}{\sin \theta} - 1} - \frac{1}{\frac{1}{\sin \theta} + 1} =$

- A. $2 \tan \theta$
- B. $2 \tan^2 \theta$
- C. $\frac{2}{\tan^2 \theta}$
- D. $\frac{2 \sin \theta}{\cos^2 \theta}$
- E. $\frac{2 \sin^2 \theta}{\cos \theta}$

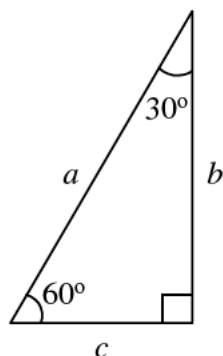
80 If $\cos \theta = x$ and $0^\circ < \theta < 90^\circ$, then $\tan \theta$
17.

- A. $\frac{1}{\sqrt{1-x^2}}$
- B. $\sqrt{1-x^2}$
- C. $\frac{\sqrt{1-x^2}}{x}$
- D. $\frac{x}{\sqrt{1-x^2}}$
- E. $\pm \frac{x}{\sqrt{1-x^2}}$

80 If $0^\circ \leq \theta < 360^\circ$, which of the following
18. equations has exactly one root?

- A. $\sin \theta = -1$
- B. $\sin \theta = -\frac{1}{2}$
- C. $\sin \theta = 0$
- D. $\sin \theta = \frac{1}{2}$
- E. $\sin \theta = 2$

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19.



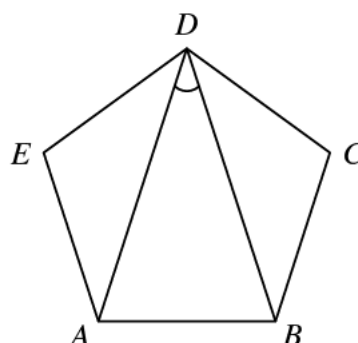
In the figure, $a : b : c =$

- A. $3 : 2 : 1$
- B. $9 : 4 : 1$
- C. $2 : \sqrt{3} : 1$
- D. $\sqrt{3} : \sqrt{2} : 1$
- E. $\sqrt{3} : 2 : 1$

80 What is the area, in cm^2 , of an
20. equilateral triangle of side x cm?

- A. $\frac{\sqrt{3}}{4}x^2$
- B. $\frac{\sqrt{3}}{2}x^2$
- C. $\frac{1}{4}x^2$
- D. $\frac{1}{2}x^2$
- E. $\sqrt{3}x^2$

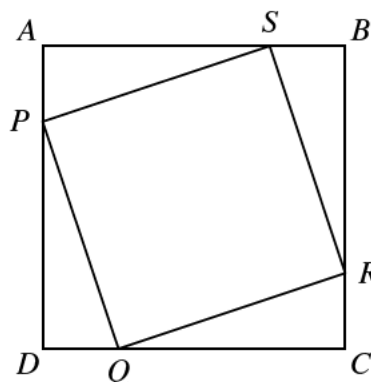
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21.



In the figure, $ABCDE$ is a regular pentagon. $\angle ADB =$

- A. 35°
- B. 36°
- C. 40°
- D. 54°
- E. 72°

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22.

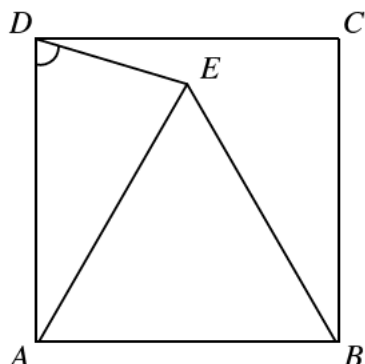


In the figure, $ABCD$ is a square with $AB = 5$. $AP = BQ = CR = DS = 1$. What is the area of $PQRS$?

- A. 9

- B. 15
C. 16
D. 17
E. 18

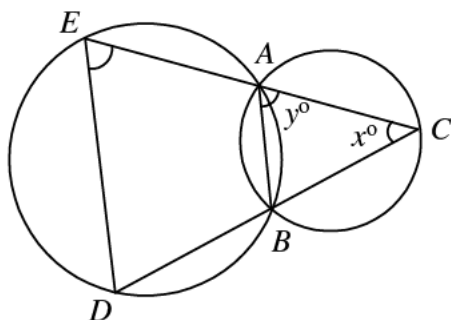
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23.



In the figure, $ABCD$ is a square and ABE is an equilateral triangle.
 $\angle ADE = ?$

- A. 72°
B. 74°
C. 76°
D. 78°
E. None of the above

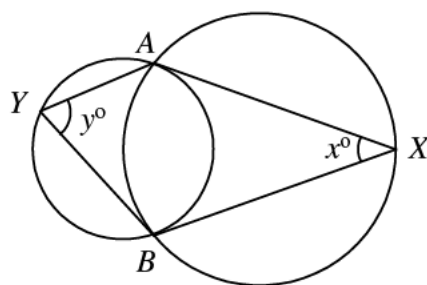
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24.



In the figure, the two circles intersect at A and B . CAE and CBD are straight lines. $\angle CED =$

- A. y°
B. $180^\circ - y^\circ$
C. $180^\circ - x^\circ - y^\circ$
D. $180^\circ - x^\circ + y^\circ$
E. $360^\circ - x^\circ - y^\circ$

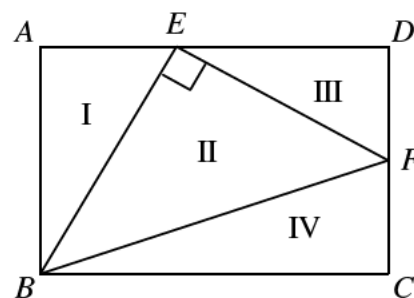
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25.



In the figure, circle AXB passes through the centre of circle AYB . $y =$

- A. $2x$
B. $180 - 2x$
C. $180 - x$
D. $\frac{1}{2}(90 - x)$
E. $\frac{1}{2}(180 - x)$

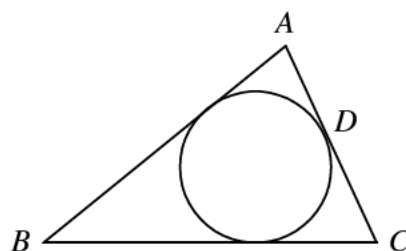
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26.



In the figure, $ABCD$ is a rectangle $\angle BEF = 90^\circ$. Which two of the triangles I, II, III and IV must be similar?

- A. I and II
B. I and III
C. II and III
D. II and IV
E. III and IV

80
27.



In the figure, the inscribed circle of $\triangle ABC$ touches AC at D . If $AB = 7$, $AC = 5$ and $AD = 2$, then $BC =$

- A. 9.5
- B. 9
- C. 8.5
- D. 8
- E. 7.5

- 80 A certain sum of money is just sufficient to pay the wages of one man for m days or the wages of one boy for n days. For how many days will this sum be just sufficient to pay the wages of one man and one boy together?

- A. $m + n$
- B. $\frac{m + n}{2}$
- C. $\frac{1}{m} + \frac{1}{n}$
- D. $\frac{m + n}{mn}$
- E. $\frac{mn}{m + n}$

- 80 If the value of $y^2 + 3y + 7$ is 2, what is the value of $2y^2 + 6y - 3$?

- A. -13
- B. -7
- C. 7
- D. 13
- E. It cannot be found from the information given

- 80 A, B, C are three spheres. If

$$\frac{\text{Surface area of } A}{\text{Surface area of } B} = 4 \text{ and } \frac{\text{Volume of } B}{\text{Volume of } C} = 2, \text{ then } \frac{\text{Volume of } A}{\text{Volume of } C} =$$

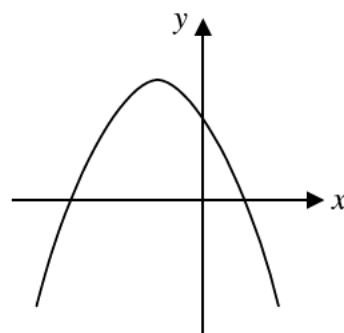
- A. 16
- B. 8
- C. 2

- D. $\frac{1}{8}$
- E. $\frac{1}{16}$

- 80 The $2n^{\text{th}}$ term of the geometric progression, 8, -4, 2, -1, ..., is

- A. $\frac{1}{2^{2n+2}}$
- B. $\frac{-1}{2^{2n+2}}$
- C. $\frac{1}{2^{2n-3}}$
- D. $\frac{1}{2^{2n-4}}$
- E. $\frac{-1}{2^{2n-4}}$

- 80
32.



The figure above shows the graph of $y = ax^2 + bx + c$. Determine whether a and c are positive or negative.

- A. $a > 0$ and $c > 0$
- B. $a < 0$ and $c < 0$
- C. $a > 0$ and $c < 0$
- D. $a < 0$ and $c > 0$
- E. It cannot be determined from the given data

- 80 \$P amounts to \$Q in n years at simple interest. The rate per annum is

- A. $\frac{100n(Q - P)}{P} \%$
- B. $\frac{100(Q - P)}{n} \%$

- C. $\frac{100(Q-P)}{nP}\%$
 D. $\frac{100(Q-P)}{nQ}\%$
 E. $100[(\frac{Q}{P})^{\frac{1}{n}} - 1]\%$

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34. If $0 < x < 1$, which of x , x^2 , $\frac{1}{x}$, $\sqrt{x}$ is the smallest? Which is the largest?

- A. $\sqrt{x}$ is smallest, x^2 is largest
 B. $\frac{1}{x}$ is smallest, x^2 is largest
 C. x is smallest, $\frac{1}{x}$ is largest
 D. x^2 is smallest, $\frac{1}{x}$ is largest
 E. x^2 is smallest, $\sqrt{x}$ is largest

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35. The Highest Common Factor of two unequal Positive integers a and b is 8. Which of the following must be true?

- I. The difference between a and b is divisible by 8
 II. $(a + b)$ is divisible by 16
 III. ab is divisible by 64
 A. III only
 B. I and II only
 C. I and III only
 D. II and III only
 E. I, II and III only

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36. x , y and z are three consecutive positive integers. Which of the following is true?

- A. $x + y + z$ must be odd
 B. $x + y + z$ must be even
 C. xyz must be odd
 D. xyz must be even
 E. $x^2 + y^2 + z^2$ must be even

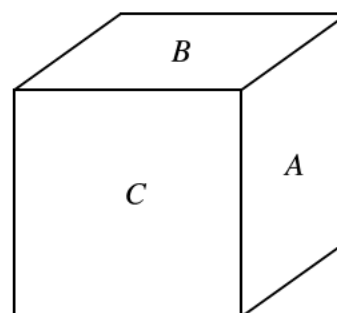
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37. If $x^2 - kx + 9 \geq 0$ for all real values of x , what is the value of k ?

- A. $k = -6$ only
 B. $k = 6$ only
 C. $-6 \leq k \leq 6$
 D. $k = 6$ or -6 only
 E. $k \leq -6$ or $k \geq 6$

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38. If x and y are real numbers, what is the minimum value of the expression $(x + y)^2 - 1$?

- A. -5
 B. -1
 C. 0
 D. 3
 E. It cannot be determined

80
39.



In the figure, the areas of the surfaces A , B , C of the cuboid are 10 cm^2 , 14 cm^2 and 35 cm^2 respectively. What is the volume of the cuboid?

- A. 49 cm^3
 B. 70 cm^3
 C. 140 cm^3
 D. 350 cm^3
 E. $4\,900 \text{ cm}^3$

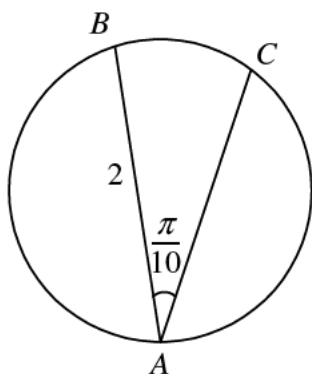
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40. x is a positive integer such that $x^2 + 2x + 7$ is even. What are the possible values of x ?

- A. x can be any positive integer
 B. x can be any positive even number
 C. x can be any positive odd number
 D. x must be an even number greater than 10 000
 E. x must be a positive odd number greater than 10 000

- 80 The perimeter of a sector is 16 and its angle is 2 radians. What is the area of the sector?

A. 16
B. 32
C. 64
D. 16π
E. 32π

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42.

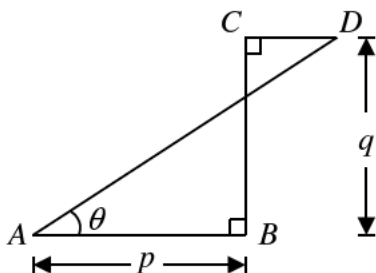


In the figure, diameter $AB = 2$.

$\angle CAB = \frac{\pi}{10}$ rad. Minor arc $BC =$

A. $\frac{\pi}{10}$
B. $\frac{\pi}{5}$
C. $\frac{3\pi}{10}$
D. $\frac{4\pi}{5}$
E. $\frac{9\pi}{10}$

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43.

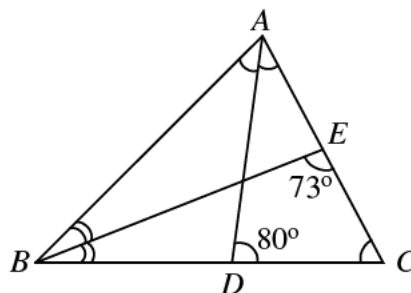


In the figure, $\angle B = \angle C = 90^\circ$.

If $AB = p$ and $BC = q$, then $CD =$

A. $p + q \tan \theta$
B. $p + \frac{q}{\tan \theta}$
C. $p + q \cos \theta$
D. $-p + q \tan \theta$
E. $-p + \frac{q}{\tan \theta}$

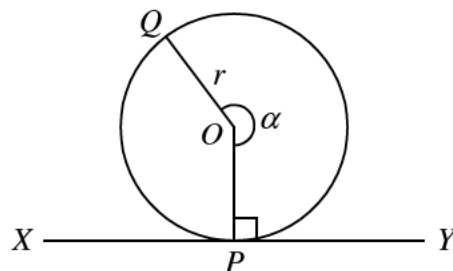
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44.



In the figure, AD and BE bisect $\angle A$ and $\angle B$ respectively. $\angle C =$

A. 50°
B. 68°
C. 74°
D. 78°
E. 80°

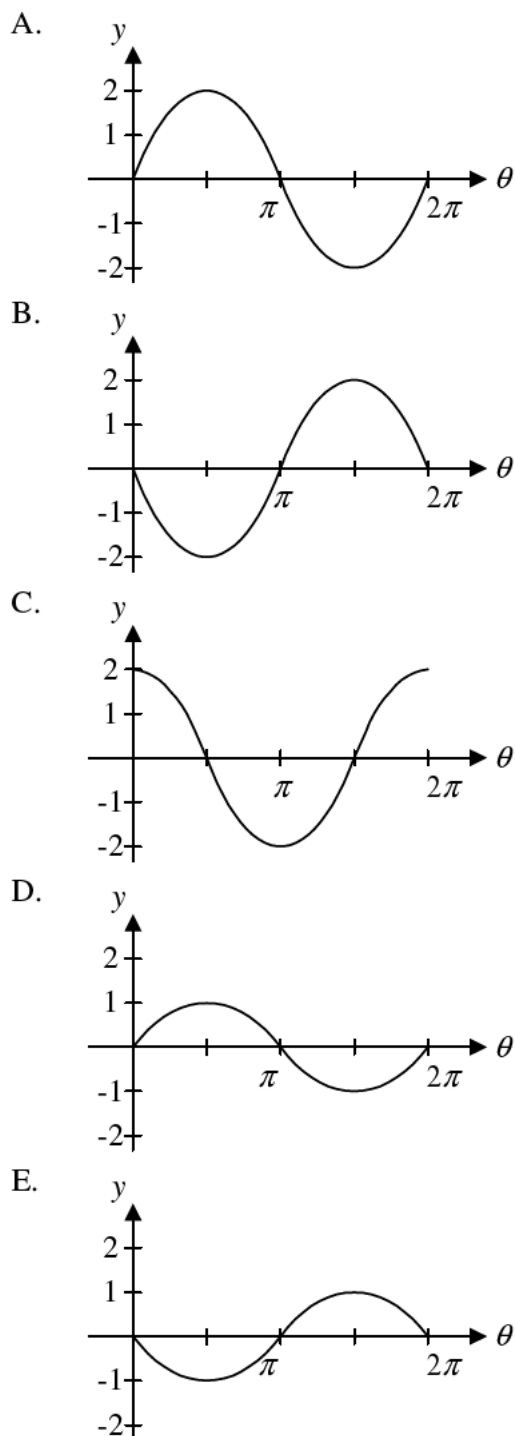
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45.



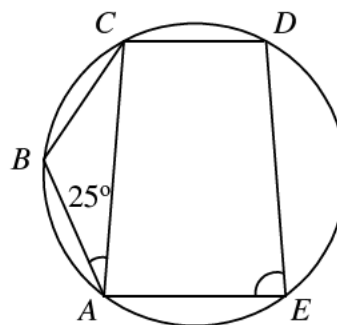
In the figure, O is the centre of the circle and its radius is r . XY touches the circle at P . Find the distance of Q from XY .

A. $r(1 - \sin \alpha)$
B. $r(1 + \sin \alpha)$
C. $r(1 - \cos \alpha)$
D. $r(1 + \cos \alpha)$
E. $r(2 - \sin \alpha)$

80 Which of the following is the graph of $y = 2 \sin \theta$, where $0 \leq \theta \leq 2\pi$?



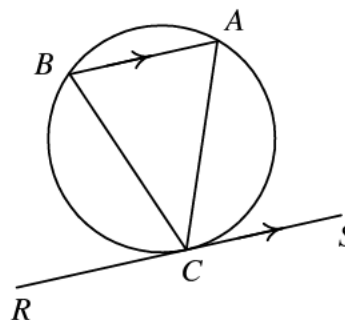
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47.



In the figure, $AB = BC = CD$. $\angle AED =$

- A. 50°
- B. 65°
- C. 75°
- D. 90°
- E. 105°

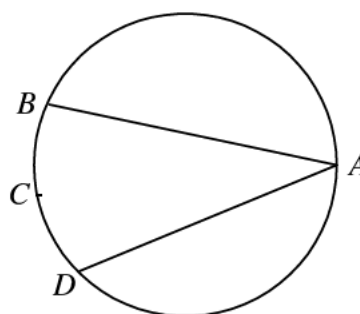
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48.



In the figure, RS is a tangent to the circle at C . BA is any chord parallel to RCS . Which of the chords AB , BC and CA must be equal in length?

- A. AB and BC only
- B. AC and BC only
- C. AB and AC only
- D. All of them
- E. No two of them

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49.

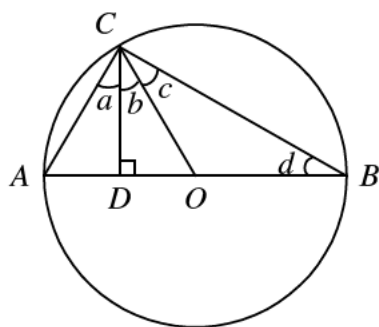


In the figure, $AB = AC$, D is the mid-point of arc BC . Which of the following is/are true?

- I. AD bisects $\angle BAC$
- II. $BC \perp AD$
- III. AD is a diameter of the circle

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

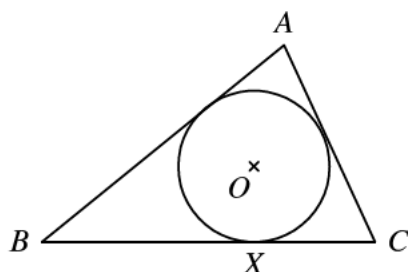
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50.



In the figure, AOB is a diameter of the circle, centre O . CD is the perpendicular bisector of OA . Which of the angles a, b, c, d is/are equal to 30° ?

- A. a only
- B. a and b only
- C. a, b and c only
- D. a, b, c and d
- E. None of them

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51.

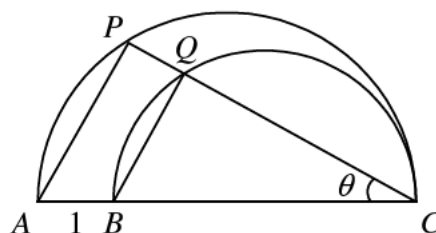


In the figure, circle O is inscribed in $\triangle ABC$, touching BC at X . Which of the following must be true?

- I. $OX \perp BC$
- II. OA bisect $\angle A$
- III. AO produced bisect BC

- A. I only
- B. I and II only
- C. I and III only
- D. I, II and III only
- E. None of them

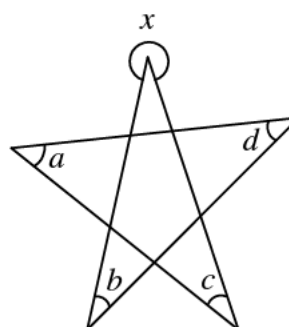
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52.



In the figure, AC and BC are diameters of two semi-circles touching each other internally at C . PQC is a straight line. If $AB = 1$, then $PQ =$

- A. $\cos \theta$
- B. $\sin \theta$
- C. $\tan \theta$
- D. $\frac{1}{\sin \theta}$
- E. $\frac{1}{\cos \theta}$

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53.

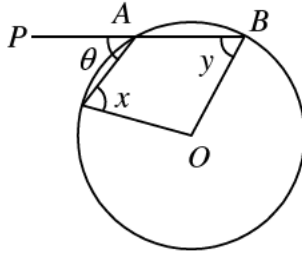


With the notation in the figure, express $a + b + c + d$ in terms of x .

- A. $x - 180^\circ$
- B. x
- C. $540^\circ - x$
- D. $360^\circ - x$

E. $180^\circ - x$

80
54.



In the figure, O is the centre of the circle. PAB is a straight line. $x + y =$

- A. 2θ
- B. $90^\circ + \theta$
- C. $180^\circ - \theta$
- D. $180^\circ - 2\theta$
- E. 180°